

Mio

Robotics and Mechanical Engineering Student

[mio0721.github.io/profile](https://github.com/mio0721) [mio0721](#)

Profile

Robotics student with hands-on experience in robot design, mechanical modeling, motion simulation, finite element analysis, embedded development, computer vision, and path planning. Experienced in leading student teams and translating engineering concepts into prototypes through 3D printing and system integration.

Education

Undergraduate Studies **Shanghai University**, School of Mechatronic Engineering and Automation

Research Projects

Industrial Pipeline Inspection Robot Based on a Halbach Magnet Array May 2023 – present
 Project Lead

- Developed the mechanical model and conducted motion simulation in Autodesk Inventor.
- Performed lightweight structural design with Altair Inspire.
- Built prototypes through 3D printing and contributed to embedded-system development.

Student Leadership

Class Monitor Dec 2021 – present
 Organized class activities and coordinated day-to-day class affairs.

Honors and Awards

17th Higher Education Cup National College Student Advanced Graphics Technology and Product Information Modeling Innovation Competition July 2024

- First Prize, Mechanical Group Competition
- Second Prize, Lightweight Design Track
- Second Prize, Advanced Graphics Technology Track
- Third Prize, Additive Manufacturing Track

China College Student Mechanical Engineering Innovation and Creativity Competition Nov 2023
 Second Prize, Regional Mechanical Product Digital Design Competition

14th Shangtu Cup Advanced Graphics Technology and Innovation Design Competition June 2024

- First Prize, Individual 3D Category
- First Prize, Individual 2D Category

Shanghai University Chuangzao Cup Competition Nov 2023
 Excellence Award

Shanghai University Honors

Apr 2024

- Top 100 Outstanding Youth League Member
- Outstanding Class Committee Member
- Outstanding Student
- Outstanding Community Individual

Shanghai University Scholarships

Jan 2024

- First-class Academic Excellence Scholarship
- Leadership Scholarship

Skills

Engineering Software: UG/NX, Autodesk Inventor, Adams, Altair Inspire**Engineering Analysis:** 3D modeling, motion simulation, finite element analysis, lightweight design**Programming and Embedded Systems:** MATLAB, Python, Raspberry Pi, Arduino**Prototyping:** 3D printing and mechanical prototype development**Languages:** English, CET-4 and CET-6